

B6 Condensed Matter Physics, 2025 — Model Solutions

Question 1 — X-ray diffraction from CaF_2

Problem. CaF_2 is fcc with Ca at $(0, 0, 0)$ and F at $\frac{1}{4}(1, 1, 1)$, $\frac{3}{4}(1, 1, 1)$. Define $S(\mathbf{G})$, derive S_{hkl} , identify the first two strong powder lines at $2\theta = 28.27^\circ, 47.00^\circ$ ($\lambda = 0.15406 \text{ nm}$), and estimate I_{111}/I_{220} .

(a) Definition and use of the structure factor

Sum of basis scatterers, weighted by atomic form factor $f_j(\mathbf{G})$ (Fourier transform of electron density, $f_j(0) = Z_j$):

$$S(\mathbf{G}) = \sum_{j \in \text{basis}} f_j(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}_j}. \quad (1)$$

Kinematic (first Born) approximation: $I(\mathbf{G}) \propto |S(\mathbf{G})|^2$ (Lorentz/polarisation/multiplicity ignored). Reflections with $S = 0$ are systematically absent.

For cubic a , $\mathbf{G}_{hkl} = (2\pi/a)(h, k, l)$. Laue $\mathbf{k}' - \mathbf{k} = \mathbf{G}$ for elastic scattering is equivalent to $\lambda = 2d_{hkl} \sin \theta$ with $d_{hkl} = 2\pi/|\mathbf{G}_{hkl}|$.

(b) Structure factor of CaF_2

Fcc Bravais factor:

$$F_{\text{fcc}}(hkl) = 1 + e^{i\pi(h+k)} + e^{i\pi(k+l)} + e^{i\pi(h+l)}, \quad (2)$$

which is 4 for h, k, l all-even or all-odd, else 0. With basis at $(0, 0, 0)$ and $\frac{1}{4}(1, 1, 1)$, $\frac{3}{4}(1, 1, 1)$, compute the phase at the F sites:

$$\mathbf{G} \cdot \frac{1}{4}(1, 1, 1)a = \frac{\pi}{2}(h + k + l).$$

Write the basis sum:

$$S_{\text{basis}} = f_{\text{Ca}} + f_{\text{F}} \left[e^{i\pi(h+k+l)/2} + e^{i3\pi(h+k+l)/2} \right].$$

Factor $e^{i\pi(h+k+l)}$ out of the F terms:

$$e^{i\pi(h+k+l)/2} + e^{i3\pi(h+k+l)/2} = e^{i\pi(h+k+l)} \left[e^{-i\pi(h+k+l)/2} + e^{i\pi(h+k+l)/2} \right].$$

Apply $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$ and note that allowed reflections (all-even or all-odd hkl) have $e^{i\pi(h+k+l)} \cos[\pi(h+k+l)/2] = \cos[\pi(h+k+l)/2]$ (the prefactor is ± 1 but cosine vanishes for all-odd):

$$S_{\text{basis}} = f_{\text{Ca}} + 2f_{\text{F}} \cos\left[\frac{\pi}{2}(h + k + l)\right].$$

Hence

$$S_{hkl} = F_{\text{fcc}}(hkl) \left[f_{\text{Ca}} + 2f_{\text{F}} \cos\left(\frac{\pi}{2}(h + k + l)\right) \right] \quad (3)$$

with $F_{\text{fcc}} = 4$ on allowed reflections.

Three regimes: $h + k + l = 4n$ gives $S \propto f_{\text{Ca}} + 2f_{\text{F}}$ (strong); $h + k + l = 4n + 2$ gives $S \propto f_{\text{Ca}} - 2f_{\text{F}}$ (near-extinct since $f_{\text{Ca}} \approx 2f_{\text{F}}$); $h + k + l$ odd gives $S \propto f_{\text{Ca}}$ (F ions cancel).

(c) Indexing the powder pattern and lattice constant

002: $h + k + l = 2$, so $S_{002} \propto f_{\text{Ca}} - 2f_{\text{F}} \approx 0$ (near-extinct). 111: $S_{111} \propto f_{\text{Ca}}$ (large).

Bragg + cubic:

$$\sin^2 \theta = \frac{\lambda^2}{4a^2} (h^2 + k^2 + l^2). \quad (4)$$

Half the scattering angles:

$$\theta_1 = 14.135^\circ, \quad \theta_2 = 23.50^\circ.$$

Compute $\sin^2 \theta$:

$$\sin^2 \theta_1 = 0.05963, \quad \sin^2 \theta_2 = 0.15903.$$

Form the ratio:

$$\frac{\sin^2 \theta_2}{\sin^2 \theta_1} = \frac{0.15903}{0.05963} \approx 2.667 \approx \frac{8}{3}.$$

Assign 111 ($h^2 + k^2 + l^2 = 3$) and 220 ($h^2 + k^2 + l^2 = 8$); the 200 ($h^2 + k^2 + l^2 = 4$) line is skipped because $h + k + l = 2$ gives the near-extinct $f_{\text{Ca}} - 2f_{\text{F}}$.

Solve Bragg for a at 111:

$$a = \frac{\lambda\sqrt{3}}{2 \sin \theta_{111}}.$$

Substitute numbers:

$$a = \frac{0.15406 \times 1.7321}{2 \times 0.24417} \text{ nm}.$$

Reduce:

$$a \approx 0.5466 \text{ nm}.$$

Cross-check from 220: $a \approx 0.5464 \text{ nm}$.

First two strong reflections: 111 and 220, $a \approx 0.547 \text{ nm}$.

(5)

(d) Form-factor graphs and intensity ratio of 111 versus 220

$f(G)$ is the Fourier transform of the ionic electron density: starts at $f(0) = Z_{\text{ion}}$ (here $f_{\text{Ca}^{2+}}(0) = 18$, $f_{\text{F}^-}(0) = 10$) and decays monotonically (closed shells).

At $a = 5.466 \text{ \AA}$:

$$G_{111} = \frac{2\pi\sqrt{3}}{a} \approx 1.99 \text{ \AA}^{-1}, \quad G_{220} = \frac{2\pi\sqrt{8}}{a} \approx 3.25 \text{ \AA}^{-1}. \quad (6)$$

Read off from the graphs at $G_{111} \approx 1.99 \text{ \AA}^{-1}$ and $G_{220} \approx 3.25 \text{ \AA}^{-1}$:

$$f_{\text{Ca}}(G_{111}) \approx 14, \quad f_{\text{F}}(G_{111}) \approx 7; \quad f_{\text{Ca}}(G_{220}) \approx 11.5, \quad f_{\text{F}}(G_{220}) \approx 5.5.$$

Drop the common $F_{\text{fcc}} = 4$. For 111, $h + k + l = 3$:

$$\cos\left(\frac{\pi}{2} \cdot 3\right) = \cos\left(\frac{3\pi}{2}\right) = 0.$$

Hence:

$$S_{111} = f_{\text{Ca}}(G_{111}) \approx 14.$$

For 220, $h + k + l = 4$:

$$\cos\left(\frac{\pi}{2} \cdot 4\right) = \cos(2\pi) = 1.$$

Hence:

$$S_{220} = f_{\text{Ca}}(G_{220}) + 2f_{\text{F}}(G_{220}) \approx 11.5 + 11 = 22.5.$$

Form the intensity ratio:

$$\frac{I_{111}}{I_{220}} = \left(\frac{S_{111}}{S_{220}} \right)^2 = \frac{14^2}{22.5^2}.$$

Reduce:

$$\frac{I_{111}}{I_{220}} = \frac{196}{506.25} \approx 0.39.$$

$$\boxed{I_{111}/I_{220} \approx 0.4.} \quad (7)$$

At 220 all three ions add coherently; at 111 the two F ions cancel and only Ca contributes.

Question 2 — 1D tight-binding model

Problem. Chain of N atoms, $\langle n | \hat{H} | n \rangle = \mathcal{E}_0$, $\langle n | \hat{H} | n \pm 1 \rangle = -t$, PBC. (a) Find $E(k)$ and $g(E)$. (b) Half-filled (monovalent) molar C and χ ; comment on divalent. (c) Add $\langle n | \hat{H} | n \pm 2 \rangle = -t/2$; modified $E(k)$, $g(E)$.

(a) Band $E(k)$, bandwidth, and density of states

Bloch ansatz $|k\rangle = N^{-1/2} \sum_n e^{ikna} |n\rangle$, PBC fixes $k = 2\pi m/(Na)$, N values in $(-\pi/a, \pi/a]$. Apply $\hat{H}$ to a single site, using only the three non-zero matrix elements:

$$\hat{H} |n\rangle = \mathcal{E}_0 |n\rangle - t |n+1\rangle - t |n-1\rangle.$$

Apply $\hat{H}$ to $|k\rangle$ and relabel the shifted sums:

$$\hat{H} |k\rangle = \frac{1}{\sqrt{N}} \sum_n e^{ikna} (\mathcal{E}_0 - te^{-ika} - te^{ika}) |n\rangle.$$

Combine the two phases via $e^{ika} + e^{-ika} = 2 \cos ka$:

$$\hat{H} |k\rangle = (\mathcal{E}_0 - 2t \cos ka) |k\rangle.$$

$$\boxed{E(k) = \mathcal{E}_0 - 2t \cos ka, \quad A = \mathcal{E}_0, \quad B = -2t.} \quad (8)$$

$$\boxed{W = 4t.} \quad (9)$$

DOS (factor 2 spin, factor 2 from $\pm k$, k -density $Na/2\pi$):

$$g(E) = \frac{2Na}{\pi} \left| \frac{dk}{dE} \right| = \frac{2Na}{\pi} \cdot \frac{1}{|2ta \sin ka|}.$$

Invert dispersion:

$$\cos ka = \frac{\mathcal{E}_0 - E}{2t}.$$

Apply $\sin^2 = 1 - \cos^2$:

$$\sin^2 ka = 1 - \left(\frac{E - \mathcal{E}_0}{2t} \right)^2.$$

Take positive root:

$$|\sin ka| = \sqrt{1 - [(E - \mathcal{E}_0)/(2t)]^2}.$$

Substitute into denominator:

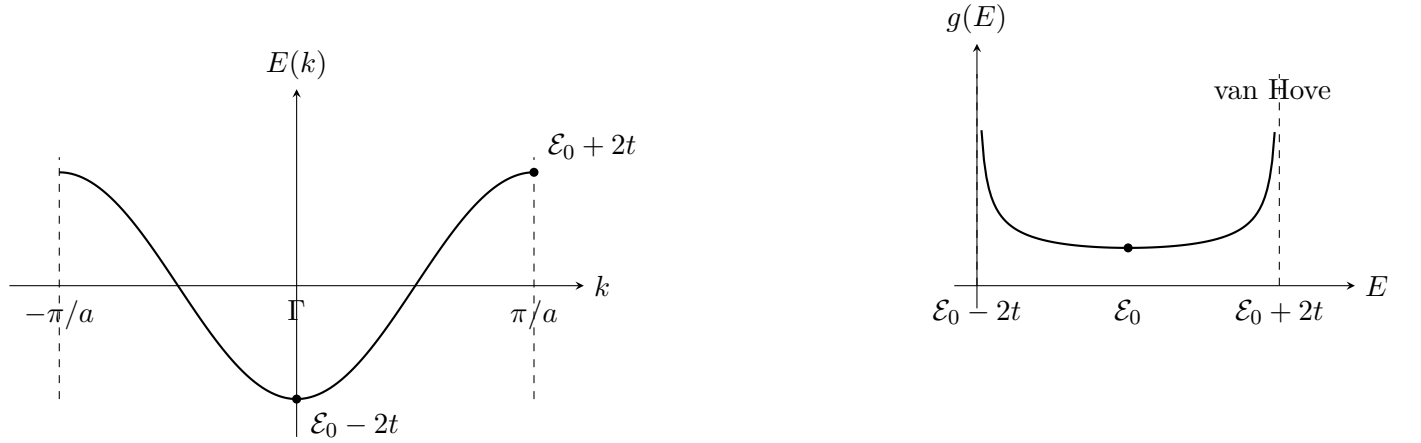
$$|2ta \sin ka| = 2ta \sqrt{1 - [(E - \mathcal{E}_0)/(2t)]^2}.$$

Combine with prefactor $2Na/\pi$:

$$g(E) = \frac{N}{\pi t} \frac{1}{\sqrt{1 - [(E - \mathcal{E}_0)/(2t)]^2}} = \frac{\alpha N}{\sqrt{1 - \beta(E - \mathcal{E}_0)^2}}, \quad (10)$$

$$\boxed{\alpha = \frac{1}{\pi t}, \quad \beta = \frac{1}{4t^2}.} \quad (11)$$

Integrable $1/\sqrt{}$ van Hove divergences at $E = \mathcal{E}_0 \pm 2t$.



(b) Heat capacity and susceptibility for a half-filled band

Monovalent: N electrons in $2N$ states $\Rightarrow$ half-filled, $E_F = \mathcal{E}_0$. From (10), $g(E_F) = N/(\pi t)$.

Sommerfeld:

$$C_V = \frac{\pi^2}{3} g(E_F) k_B^2 T.$$

Substitute $g(E_F) = N/(\pi t)$:

$$C_V = \frac{\pi^2}{3} \frac{N}{\pi t} k_B^2 T.$$

Cancel π :

$$C_V = \frac{\pi}{3} N k_B \frac{k_B T}{t}.$$

Per mole ($N = N_A$):

$$C = \frac{\pi}{3} R \frac{k_B T}{t}, \quad \boxed{\xi = 1, \quad \eta = \pi/3.} \quad (12)$$

Pauli (volume) susceptibility:

$$\chi_P = \frac{\mu_0 \mu_B^2 g(E_F)}{V}.$$

Substitute $g(E_F) = N/(\pi t)$ and write the molar form ($N \rightarrow N_A$):

$$\chi = \mu_0 \mu_B^2 \cdot \frac{N_A}{\pi t}.$$

Use $N_A = R/k_B$:

$$\chi = \mu_0 \mu_B^2 \frac{N_A}{\pi t} = \frac{\mu_0 \mu_B^2 R}{\pi t k_B}. \quad (13)$$

Temperature-independent.

Divalent. $2N$ electrons fill the band; μ in gap above. Insulator: $C_{\text{el}} \propto e^{-E_g/k_B T}$ (phonon T^3 dominates), $\chi_P \rightarrow 0$ (only small Landau/core diamagnetism).

(c) Adding second-neighbour hopping

Acting on $|k\rangle$, the second-neighbour terms contribute $-(t/2)(e^{-2ika} + e^{2ika})$. Combine into a cosine:

$$-\frac{t}{2}(e^{-2ika} + e^{2ika}) = -t \cos(2ka).$$

Add to the first-neighbour dispersion:

$$E(k) = \mathcal{E}_0 - 2t \cos ka - t \cos(2ka). \quad (14)$$

Differentiate:

$$\frac{dE}{dk} = 2ta \sin ka + 2ta \sin(2ka).$$

Factor using $\sin(2ka) = 2 \sin ka \cos ka$:

$$\frac{dE}{dk} = 2ta \sin ka (1 + 2 \cos ka).$$

Set to zero:

$$\sin ka = 0 \quad \text{or} \quad \cos ka = -\frac{1}{2} \Rightarrow ka = 0, \pm \frac{2\pi}{3}, \pm \pi.$$

Evaluate at $ka = 0$:

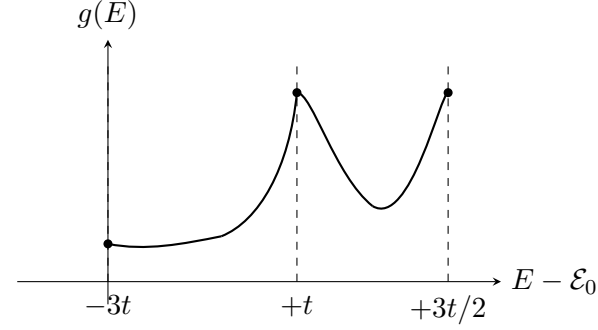
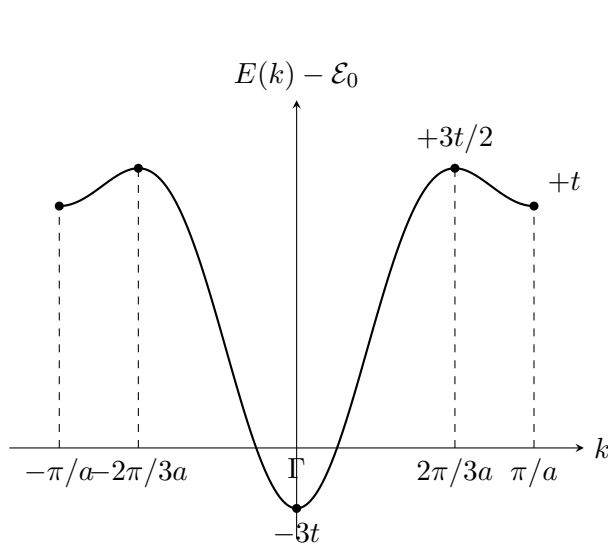
$$E(0) = \mathcal{E}_0 - 2t - t = \mathcal{E}_0 - 3t \quad (\text{global min}).$$

Evaluate at $ka = \pm 2\pi/3$, using $\cos(2\pi/3) = -1/2$ and $\cos(4\pi/3) = -1/2$:

$$E\left(\pm \frac{2\pi}{3a}\right) = \mathcal{E}_0 - 2t\left(-\frac{1}{2}\right) - t\left(-\frac{1}{2}\right) = \mathcal{E}_0 + \frac{3t}{2} \quad (\text{global max}).$$

Evaluate at $ka = \pm \pi$:

$$E\left(\pm \frac{\pi}{a}\right) = \mathcal{E}_0 + 2t - t = \mathcal{E}_0 + t \quad (\text{local min}).$$



Three van Hove singularities, at $E = E_0 - 3t, +t, +3t/2$.

Question 3 — Semiconductors and the Hall effect

Problem. Define intrinsic/extrinsic semiconductors; derive $n(T) \propto T^\alpha e^{-\eta E_g/k_B T}$ and give $p(T)$. Define and measure R_H ; use $R_H(300)/R_H(350) = 28$ in undoped Si to get E_g . Estimate donor Bohr radius and ionisation energy in P-doped Si ($\epsilon_r = 11.7$, $m^* = 1.09 m_e$); sketch $\ln R_H$ vs $1/T$.

(a) Intrinsic and extrinsic semiconductors

Intrinsic: pure host, valence full / conduction empty at $T = 0$, gap E_g . Thermal pair excitation gives $n = p$. *Extrinsic:* dopants put levels in the gap. Donors (e.g. P in Si) sit $E_d \ll E_g$ below E_c and donate electrons; acceptors sit above E_v and accept (creating holes). $n \neq p$ in general; majority carriers come from impurities.

(b) Conduction-band electron density $n(T)$

Parabolic band:

$$g_c(E) = \frac{1}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}. \quad (15)$$

Non-degenerate (μ in gap): use Maxwell–Boltzmann tail. With $x = (E - E_c)/k_B T$ and $\int_0^\infty \sqrt{x} e^{-x} dx = \sqrt{\pi}/2$,

$$n(T) = N_c(T) e^{-(E_c - \mu)/k_B T}, \quad N_c(T) = 2 \left(\frac{m_e^* k_B T}{2\pi \hbar^2} \right)^{3/2}. \quad (16)$$

Analogously for holes:

$$p(T) = N_v(T) e^{-(\mu - E_v)/k_B T}, \quad N_v(T) = 2 \left(\frac{m_h^* k_B T}{2\pi \hbar^2} \right)^{3/2}. \quad (17)$$

Multiply n and p :

$$np = N_c N_v e^{-(E_c - E_v)/k_B T}.$$

Use $E_c - E_v = E_g$ and $N_c N_v \propto T^3$:

$$np = N_c N_v e^{-E_g/k_B T} \propto T^3 e^{-E_g/k_B T}.$$

Impose intrinsic neutrality $n = p$:

$$n^2 = N_c N_v e^{-E_g/k_B T}.$$

Take square root:

$$n(T) = p(T) = \sqrt{N_c N_v} e^{-E_g/(2k_B T)} \propto T^{3/2} e^{-E_g/(2k_B T)}.$$

$$\boxed{\alpha = 3/2, \quad \eta = 1/2.} \quad (18)$$

$$\boxed{p(T) = n(T) \propto T^{3/2} e^{-E_g/(2k_B T)}.} \quad (19)$$

The $E_g/2$ in the Boltzmann factor reflects μ pinned mid-gap.

(c) Hall coefficient: definition, measurement, and E_g from undoped Si

Definition. With j_x , B_z , transverse field E_y balances the Lorentz force in steady state:

$$R_H = \frac{E_y}{j_x B_z}. \quad (20)$$

Single carrier of charge q , density n :

$$\boxed{R_H = \frac{1}{nq}.} \quad (21)$$

Sign distinguishes electrons ($R_H < 0$) from holes ($R_H > 0$).

Measurement. Thin rectangular bar; force longitudinal current I , measure transverse Hall voltage V_H with high-impedance voltmeter. Width w , thickness d : $R_H = V_H d / (IB)$. Reverse B and average to remove offsets.

Why electrons dominate in Si. $\mu_e \approx 1400 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1} \gg \mu_h \approx 450 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$; in two-carrier intrinsic Si, $R_H = (p\mu_h^2 - n\mu_e^2) / [e(n\mu_e + p\mu_h)^2] < 0$, electron-dominated.

Extracting E_g . Use $R_H \propto 1/n$ with $n \propto T^{3/2} e^{-E_g/(2k_B T)}$:

$$\frac{R_H(300)}{R_H(350)} = \left(\frac{350}{300} \right)^{3/2} \exp \left[\frac{E_g}{2k_B} \left(\frac{1}{300} - \frac{1}{350} \right) \right] = 28. \quad (22)$$

Evaluate the prefactor:

$$\left(\frac{350}{300} \right)^{3/2} \approx 1.260.$$

Isolate the exponential:

$$\exp \left[\frac{E_g}{2k_B} \left(\frac{1}{300} - \frac{1}{350} \right) \right] = \frac{28}{1.260} \approx 22.2.$$

Take the logarithm:

$$\frac{E_g}{2k_B} \left(\frac{1}{300} - \frac{1}{350} \right) = \ln 22.2 \approx 3.10.$$

Evaluate the temperature factor:

$$\frac{1}{300} - \frac{1}{350} = 4.762 \times 10^{-4} \text{ K}^{-1}.$$

Solve for E_g :

$$E_g = \frac{2k_B \times 3.10}{4.762 \times 10^{-4}} \text{ J} \approx 1.80 \times 10^{-19} \text{ J}.$$

$$\boxed{E_g \approx 1.12 \text{ eV}.} \quad (23)$$

(d) Donor Bohr radius and ionisation energy in P-doped Si

Hydrogenic in dielectric medium with effective mass: $\epsilon_0 \rightarrow \epsilon_0 \epsilon_r$, $m_e \rightarrow m_e^*$:

$$a^* = \epsilon_r \frac{m_e}{m_e^*} a_0, \quad E_R^* = \frac{1}{\epsilon_r^2} \frac{m_e^*}{m_e} E_R. \quad (24)$$

With $\epsilon_r = 11.7$, $m_e^*/m_e = 1.09$, $a_0 = 0.529 \text{ \AA}$, $E_R = 13.6 \text{ eV}$, substitute into a^* :

$$a^* = \frac{11.7}{1.09} \times 0.529 \text{ \AA}.$$

Reduce the fraction:

$$a^* = 10.73 \times 0.529 \text{ \AA} \approx 5.68 \text{ \AA}.$$

Substitute into E_R^* :

$$E_R^* = \frac{1.09}{11.7^2} \times 13.6 \text{ eV}.$$

Reduce:

$$E_R^* = \frac{1.09}{136.89} \times 13.6 \text{ eV} \approx 0.108 \text{ eV}.$$

$$\boxed{a^* \approx 0.57 \text{ nm}, \quad E_d \approx 0.11 \text{ eV}.} \quad (25)$$

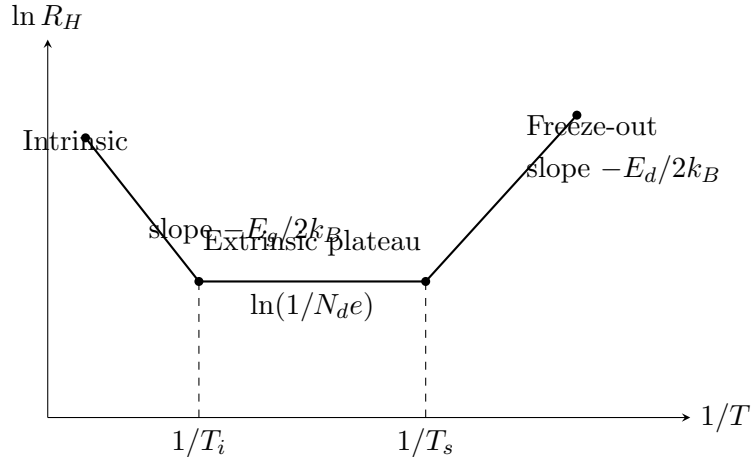
$a^* \gg a_{\text{Si}}$ justifies the dielectric continuum picture; experimental $E_d \approx 45 \text{ meV}$ for P:Si, the discrepancy from anisotropic mass and central-cell corrections.

(e) Sketch of $\ln R_H$ versus $1/T$ for the P-doped sample

Three regimes (rising T):

1. Freeze-out ($k_B T \ll E_d$): $n \propto e^{-E_d/(2k_B T)}$, $\ln R_H$ rises with $1/T$, slope $\sim -E_d/2k_B$.
2. Saturation: all donors ionised, $n \approx N_d$, $\ln R_H$ flat.
3. Intrinsic: $n_i > N_d$, $\ln R_H \propto E_g/(2k_B T)$, steep slope $-E_g/2k_B$.

Crossover when $\sqrt{N_c N_v} e^{-E_g/(2k_B T)} = N_d$; for $N_d = 6.25 \times 10^{18} \text{ m}^{-3}$, $E_g = 1.12 \text{ eV}$, $T \sim 500 \text{ K}$.



Question 4 — Heat capacity, Debye and magnetic contributions

Problem. Compare Einstein/Debye assumptions; derive low- T $C_{\text{Debye}} = xpR(T/T_D)^y$. For N non-interacting $S = 1/2$ spins derive $Z = [2 \cosh z]^N$, hence χ and C_{mag} . Compare $C_{\text{mag}}/C_{\text{Debye}}$ at 10 K, $B = 0.1$ T and 10 T, with $T_D = 200$ K, $p = 10$, one magnetic atom per formula unit.

(a) Einstein versus Debye models

Both: harmonic phonons, independent modes, Bose–Einstein occupation, Dulong–Petit limit $3Nk_B$ at high T .

Differ: Einstein gives all $3N$ modes $\omega = \omega_E$ (good for optical, zone-centre); $C \propto e^{-\hbar\omega_E/k_B T}$ at low T — too rapid. Debye uses linear acoustic dispersion $\omega = c_s k$ up to cutoff ω_D (mode count = $3N$); captures long-wavelength modes correctly, gives $C \propto T^3$.

(b) Debye internal energy and low- T heat capacity

Debye DOS (3 acoustic branches, N_a atoms):

$$g(\omega) = \frac{9N_a}{\omega_D^3} \omega^2, \quad 0 \leq \omega \leq \omega_D. \quad (26)$$

$$U(T) = \int_0^{\omega_D} d\omega g(\omega) \frac{\hbar\omega}{e^{\hbar\omega/k_B T} - 1}.$$

Substitute $x = \hbar\omega/k_B T$, $T_D = \hbar\omega_D/k_B$:

$$U(T) = 9N_a k_B T \left(\frac{T}{T_D} \right)^3 \int_0^{T_D/T} \frac{x^3 dx}{e^x - 1}.$$

Low T , extend the upper limit to ∞ :

$$U(T) \approx 9N_a k_B T \left(\frac{T}{T_D} \right)^3 \int_0^{\infty} \frac{x^3 dx}{e^x - 1}.$$

Use $\int_0^{\infty} x^3/(e^x - 1) dx = \pi^4/15$:

$$U(T) \approx 9N_a k_B T \left(\frac{T}{T_D} \right)^3 \cdot \frac{\pi^4}{15}.$$

Simplify:

$$U(T) \approx \frac{3\pi^4}{5} N_a k_B T \left(\frac{T}{T_D} \right)^3.$$

Differentiate with respect to T :

$$C_V = \frac{12\pi^4}{5} N_a k_B \left(\frac{T}{T_D} \right)^3.$$

Per mole of formula units, $N_a = pN_A$:

$$\boxed{C_{\text{Debye}}(T) = \frac{12\pi^4}{5} pR \left(\frac{T}{T_D} \right)^3, \quad x = \frac{12\pi^4}{5}, \quad y = 3, \quad T_D = \frac{\hbar\omega_D}{k_B}.} \quad (27)$$

(c) Paramagnet: partition function, susceptibility, and C_{mag}

$S = 1/2$, $g_s = 2$: $\epsilon_{\pm} = \mp \mu_B B$, so

$$Z_1 = 2 \cosh z, \quad z \equiv \frac{\mu_B B}{k_B T}.$$

Independent spins:

$$Z = [2 \cosh(z)]^N. \quad (28)$$

Free energy:

$$F = -N k_B T \ln(2 \cosh z).$$

Differentiate w.r.t. B using $\partial z / \partial B = \mu_B / (k_B T)$:

$$\frac{\partial F}{\partial B} = -N k_B T \tanh z \cdot \frac{\mu_B}{k_B T} = -N \mu_B \tanh z.$$

Magnetisation per unit volume:

$$M = -\frac{1}{V} \frac{\partial F}{\partial B} = \frac{N \mu_B}{V} \tanh z.$$

Curie limit $z \ll 1$, expand $\tanh z \approx z$:

$$M \approx \frac{N \mu_B}{V} z.$$

Substitute $z = \mu_B B / (k_B T)$:

$$M \approx \frac{N \mu_B^2 B}{V k_B T}.$$

Differentiate w.r.t. B at $B = 0$:

$$\chi(T) = \mu_0 \left. \frac{\partial M}{\partial B} \right|_{B=0} = \frac{\mu_0 N \mu_B^2}{V k_B T}. \quad (29)$$

Internal energy from $U = -\partial \ln Z / \partial \beta$ with $\beta = 1 / (k_B T)$:

$$U_{\text{mag}} = -N \frac{\partial}{\partial \beta} \ln[2 \cosh(\mu_B B \beta)] = -N \mu_B B \tanh(\mu_B B \beta).$$

Rewrite in terms of $z = \mu_B B \beta$:

$$U_{\text{mag}} = -N \mu_B B \tanh z.$$

Use $dz/dT = -z/T$ and $d(\tanh z)/dz = \text{sech}^2 z$:

$$\frac{dU_{\text{mag}}}{dT} = -N \mu_B B \text{sech}^2 z \cdot \left(-\frac{z}{T}\right).$$

Cancel the two minus signs:

$$C_{\text{mag}} = \frac{dU_{\text{mag}}}{dT} = N \mu_B B \text{sech}^2 z \cdot \frac{z}{T}.$$

Substitute $\mu_B B = z k_B T$:

$$C_{\text{mag}} = N k_B z^2 \text{sech}^2 z.$$

Rewrite $\text{sech}^2 z = 1 / \cosh^2 z$ and pass to a mole of spins:

$$\boxed{C_{\text{mag}}(T, B) = R [f(z)]^2, \quad f(z) = \frac{z}{\cosh z}, \quad z = \frac{\mu_B B}{k_B T}. \quad (30)}$$

Schottky form: $\propto z^2$ for $z \ll 1$, exponentially suppressed for $z \gg 1$, peak near $z \approx 1.2$.

(d) Numerical comparison and effect of strong interactions

At $T = 10$ K, write the Debye contribution:

$$C_{\text{Debye}} = \frac{12\pi^4}{5} \cdot p \cdot R \left(\frac{T}{T_D} \right)^3.$$

Substitute $p = 10$, $T = 10$ K, $T_D = 200$ K:

$$C_{\text{Debye}} = \frac{12\pi^4}{5} \cdot 10 \cdot R \left(\frac{10}{200} \right)^3.$$

Evaluate the prefactor:

$$\frac{12\pi^4}{5} \approx 233.78.$$

Evaluate the cube:

$$\left(\frac{10}{200} \right)^3 = \frac{1}{8000} = 1.25 \times 10^{-4}.$$

Multiply:

$$C_{\text{Debye}} \approx 233.78 \times 10 \times 1.25 \times 10^{-4} R \approx 0.2922 R.$$

One mole of spins per mole of formula unit. With $\mu_B/k_B = 0.6717$ K T⁻¹:

(i) $B = 0.1$ T: evaluate z :

$$z = \frac{\mu_B B}{k_B T} = \frac{0.6717 \times 0.1}{10} \approx 6.72 \times 10^{-3}.$$

In the Curie limit $z \ll 1$, $\cosh z \approx 1$:

$$C_{\text{mag}} \approx R z^2.$$

Square z :

$$C_{\text{mag}} \approx R \times (6.72 \times 10^{-3})^2 \approx 4.51 \times 10^{-5} R.$$

Divide by $C_{\text{Debye}} = 0.2922 R$:

$$\frac{C_{\text{mag}}}{C_{\text{Debye}}} \approx \frac{4.51 \times 10^{-5}}{0.2922} \approx 1.54 \times 10^{-4}.$$

$$\boxed{\left. \frac{C_{\text{mag}}}{C_{\text{Debye}}} \right|_{0.1 \text{ T}} \approx 1.5 \times 10^{-4}.} \quad (31)$$

(ii) $B = 10$ T: evaluate z :

$$z = \frac{0.6717 \times 10}{10} = 0.6717.$$

Evaluate $\cosh z$:

$$\cosh z \approx 1.232.$$

Form $f(z) = z / \cosh z$:

$$f(z) \approx \frac{0.6717}{1.232} \approx 0.5453.$$

Square and multiply by R :

$$C_{\text{mag}} = R [f(z)]^2 \approx 0.2974 R.$$

Divide by $C_{\text{Debye}} = 0.2922 R$:

$$\frac{C_{\text{mag}}}{C_{\text{Debye}}} \approx \frac{0.2974}{0.2922} \approx 1.02.$$

$$\boxed{\left. \frac{C_{\text{mag}}}{C_{\text{Debye}}} \right|_{10 \text{ T}}} \approx 1. \quad (32)$$

At 10 T the Schottky term is comparable to the phonon background; cooling a magnetic compound by adiabatic demagnetisation exploits exactly this competition.

Strongly interacting spins at $B = 0$. Exchange J generates an internal field; below $T_c \sim J/k_B$ the spins order. Expect:

- Lambda peak in $C_{\text{mag}}(T, 0)$ at T_c (second-order transition).
- Magnon power law below T_c : $T^{3/2}$ (3D ferromagnet, $\omega \propto k^2$); T^3 (3D antiferromagnet, $\omega \propto k$).
- High- T tail $\propto 1/T^2$ from short-range correlations.
- Total entropy $\int (C_{\text{mag}}/T) dT = R \ln 2$ conserved (spin degeneracy).
- Curie–Weiss susceptibility $\chi \propto 1/(T - \Theta_W)$, $\Theta_W \gtrless 0$ for FM/AFM.